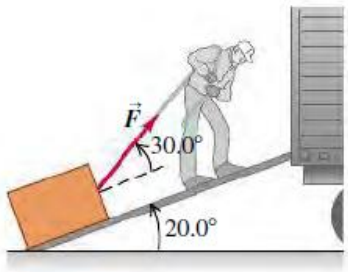
COM105 Engineering Physics (I)

Tutorial 4

1. Two dogs pull on ropes attached to a post. Dog A exerts a force of 270 N horizontally and dog B exerts a force of 300 N. If the angle between the ropes is 60.0°, find the magnitude of the resultant force and the angle it makes with Rover's rope.

2. A man is dragging a trunk up the loading ramp of a mover's truck. The ramp has a slope angle of 20.0°, and the man pulls upward with a force **F** whose direction makes an angle of 30.0° with the ramp (Fig. E4.4). (a) How large a force **F** is necessary for the component F_x parallel to the ramp to be 90.0 N? (b) How large will the component F_y perpendicular to the ramp be then?

Figure **E4.4**



3. An electron (mass = 9.11×10^{-31} kg) leaves one end of a TV picture tube with zero initial speed and travels in a straight line to the accelerating grid, which is 1.80 cm away. It reaches the grid with a speed of 3.00×10^6 m/s. If the accelerating force is constant, compute (a) the acceleration; (b) the time to reach the grid; and (c) the net force, in newtons. Ignore the gravitational force on the electron.

4. A hockey puck with mass 0.160 kg is at rest at the origin (x = 0) on the horizontal, frictionless surface of the rink. At time t = 0 a player applies a force of 0.250 N to the puck, parallel to the x-axis; she continues to apply this force until t = 2.00 s. (a) What are the position and speed of the puck at t = 2.00 s? (b) If the same force is again applied at t = 5.00 s, what are the position and speed of the puck at t = 7.00 s?

5. A small 8.00-kg rocket burns fuel that exerts a time varying upward force on the rocket (assume constant mass) as the rocket moves upward from the launch pad. This force obeys the equation $F = A + Bt^2$. Measurements show that at $t = 0$, the force is 100.0 N, and at the end of the first 2.00 s, it is 150.0 N. (a) Find the constants A and B, including their SI units. (b) Find the net force on this rocket and its acceleration (i) the instant after the fuel ignites and (ii) 3.00 s after the fuel ignites. (c) Suppose that you were using this rocket in outer space, far from all gravity. What would its acceleration be 3.00 s after fuel ignition?

6. The upward normal force exerted by the floor is 620 N on an elevator passenger who weighs 650 N. What are the reaction forces to these two forces? Is the passenger accelerating? If so, what are the magnitude and direction of the acceleration?

7. A ball is hanging from a long string that is tied to the ceiling of a train car traveling eastward on horizontal tracks. An observer inside the train car sees the ball hang motionless. Draw a clearly labeled free-body diagram for the ball if (a) the train has a uniform velocity and (b) the train is speeding up uniformly. Is the net force on the ball zero in either case? Explain.

8. The fastest served tennis ball, served by “Big Bill” Tilden in 1931, was measured at 73.14 m/s. The mass of a tennis ball is 57 g, and the ball, which starts from rest, is typically in contact with the tennis racquet for 30.0 ms. Assuming constant acceleration, (a) what force did Big Bill’s tennis racquet exert on the ball if he hit it essentially horizontally? (b) Draw free-body diagrams of the ball during the serve and just after it moved free of the racquet.